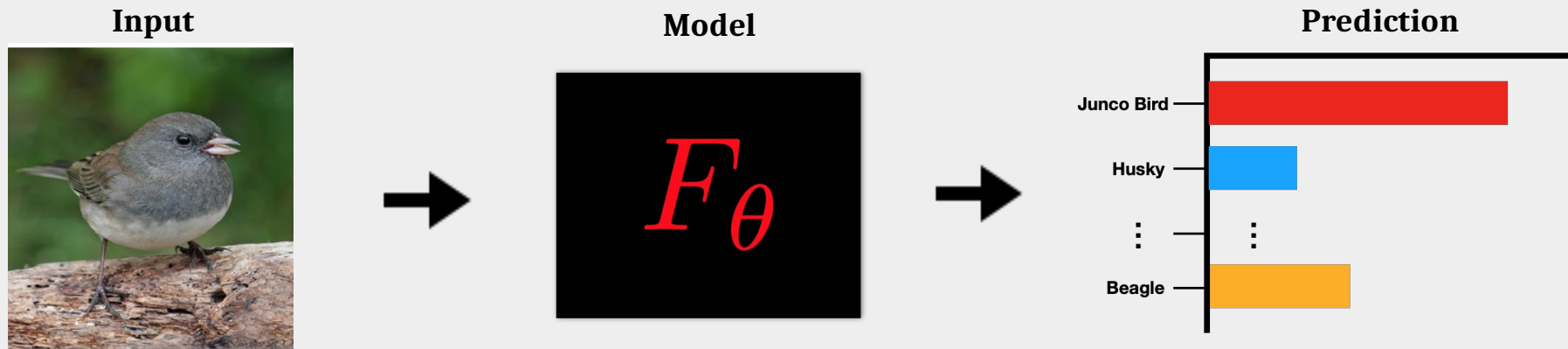


Sanity Checks for Saliency Maps

Overview

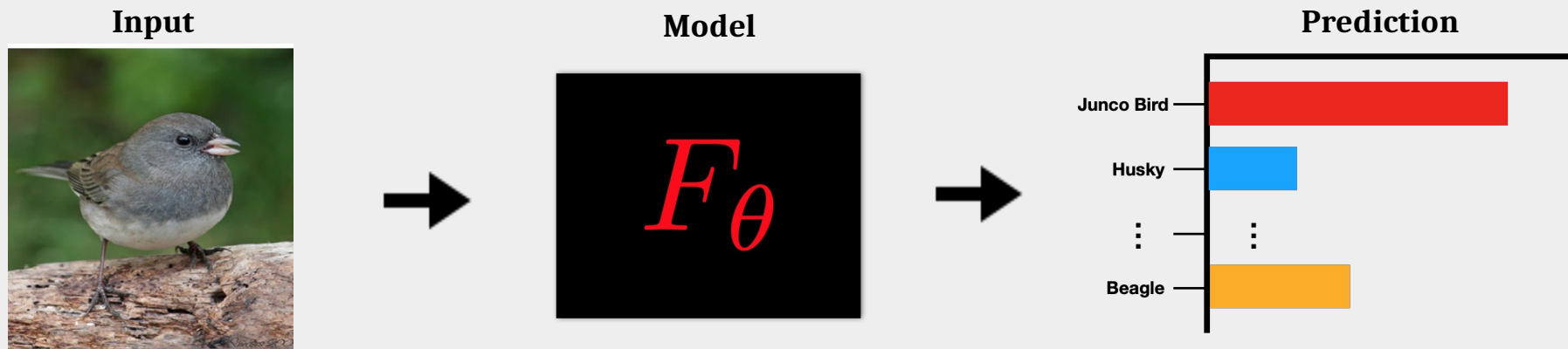
- Feature Attribution / Saliency Maps Setup
- Overview of Sanity Checks for Saliency Maps
- Follow-up work
- Parting thoughts / Q&A

Feature Attributions / Saliency Maps

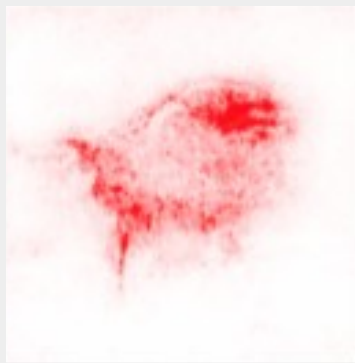


What parts of the input are 'most important' for the model prediction Junco Bird?

Feature Attributions / Saliency Maps

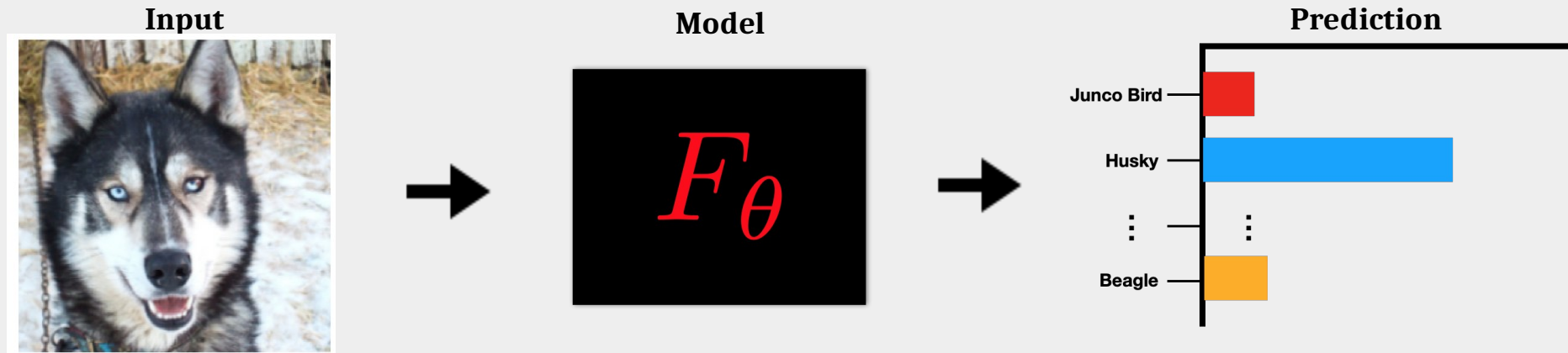


What parts of the input are 'most important' for the model prediction Junco Bird?



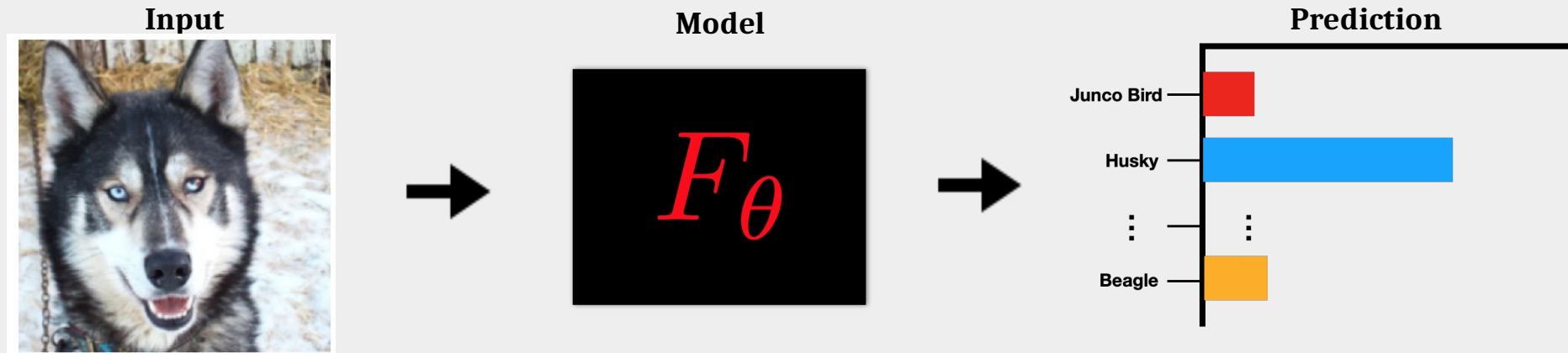
Feature Attribution / Saliency Map

Identifying Shortcuts

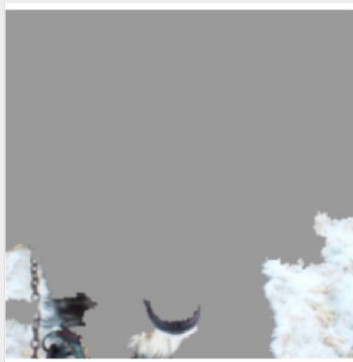


What parts of the input are ‘most important’ for the model prediction **Husky**?

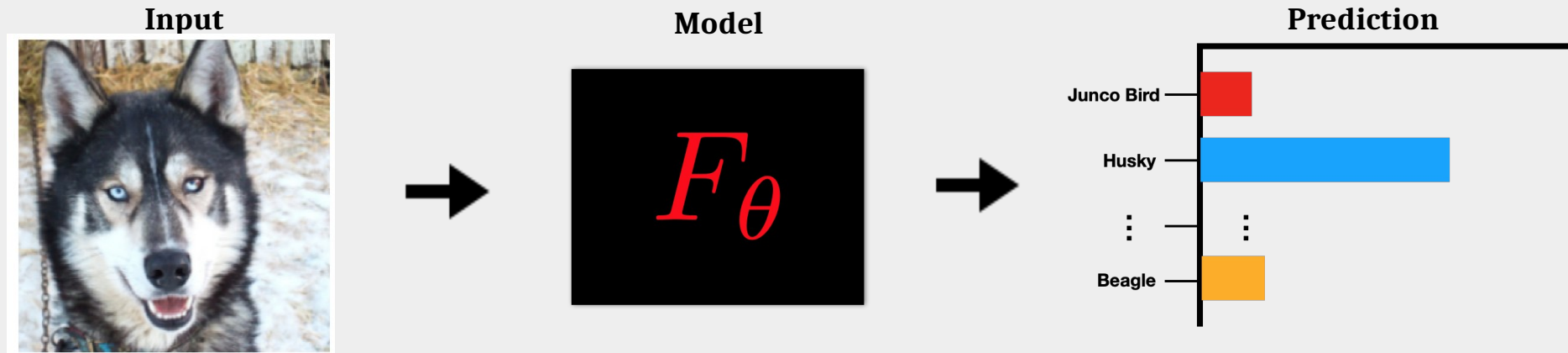
Identifying Shortcuts



What parts of the input are 'most important' for the model prediction **Husky**?

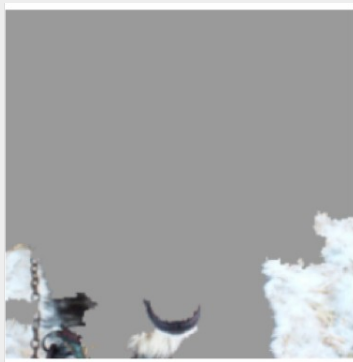


Identifying Shortcuts



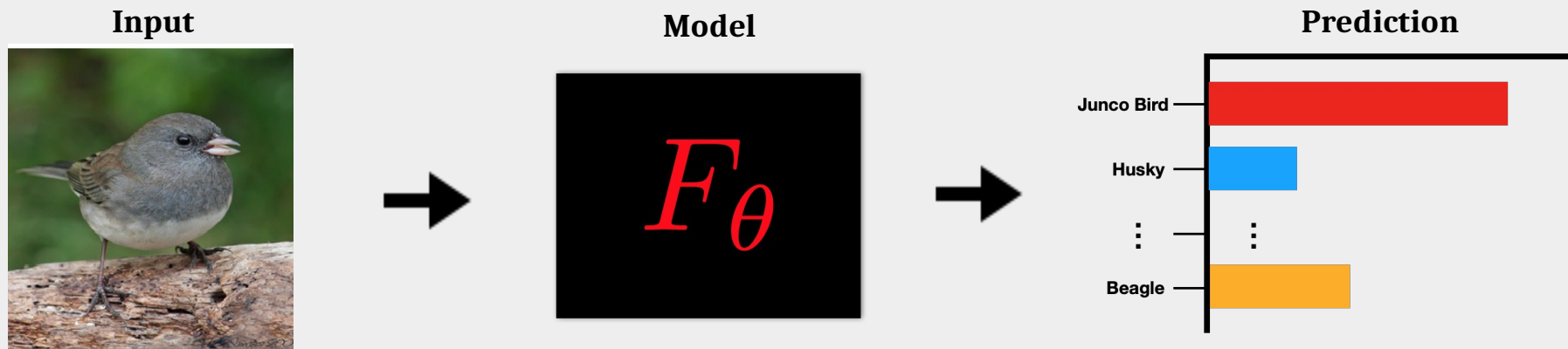
What parts of the input are 'most important' for the model prediction **Husky**?

Collect additional data
to fix the bug.



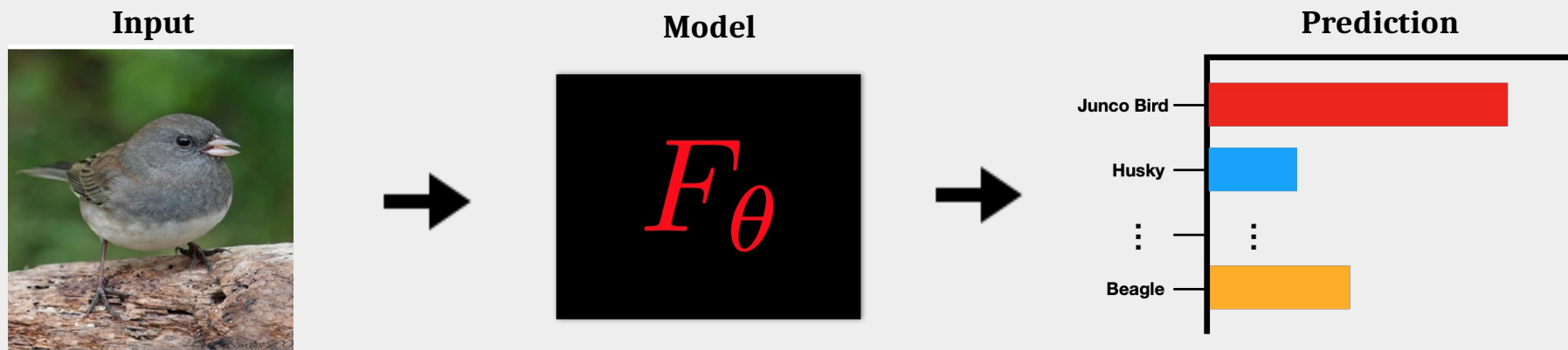
Model relying on snow to identify Huskies.

Feature Attributions / Saliency Maps



Feature attribution method: assigns an output 'relevance' score to each dimension of the input.

Feature Attributions / Saliency Maps



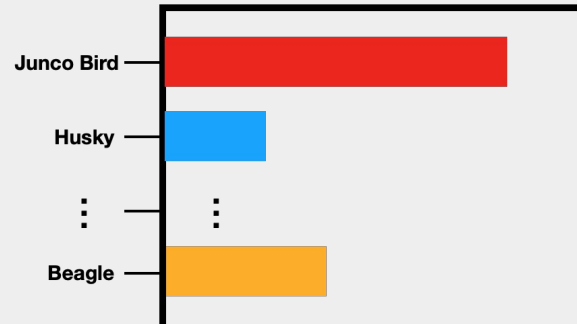
Feature attribution method: assigns an output 'relevance' score to each dimension of the input.

$$\begin{array}{c} \text{Model} \\ \hline F : \mathbb{R}^d \rightarrow \mathbb{R}^c \\ \hline F_i : \mathbb{R}^d \rightarrow \mathbb{R} \quad \text{class specific logit} \end{array}$$

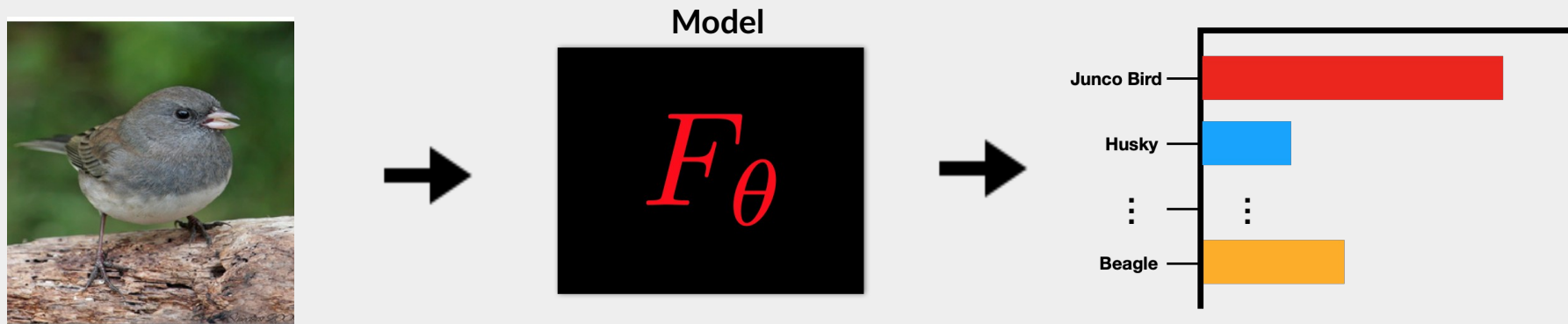
Input-Gradient / Saliency / Gradient



Model



Input-Gradient / Saliency / Gradient



Input-Gradient

$$\nabla_x F_i(x) \rightarrow \in \mathbb{R}^d$$

Input

Logit

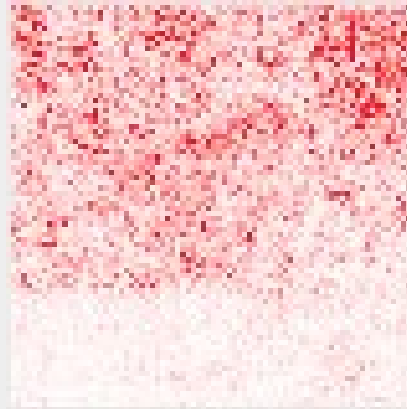
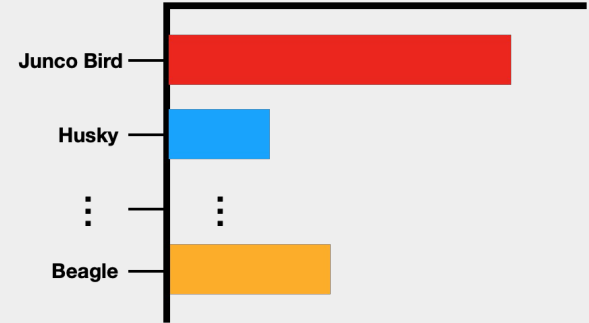
Same dimension as the input.

[Baehrens et. al. 2010](#); [Simonyan et. al. 2014](#)

Input-Gradient / Saliency / Gradient



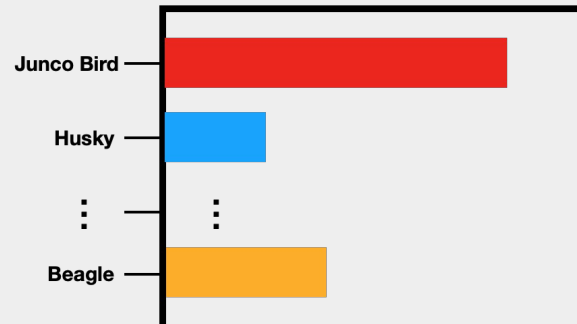
Model



Integrated Gradients



Model



$$(x - \tilde{x}) \times \int_{\alpha=0}^1 \frac{\partial F(\tilde{x} + \alpha \times (x - \tilde{x}))}{\partial x}$$



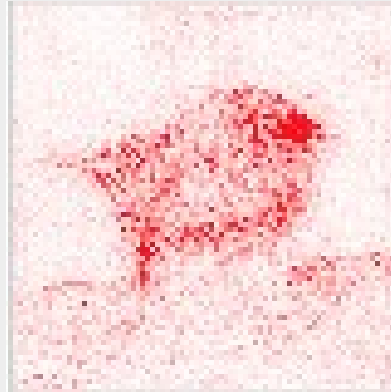
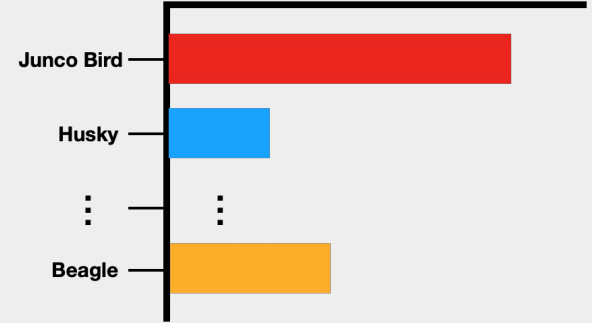
Baseline input

Path integral: 'sum' of interpolated gradients

Integrated Gradients



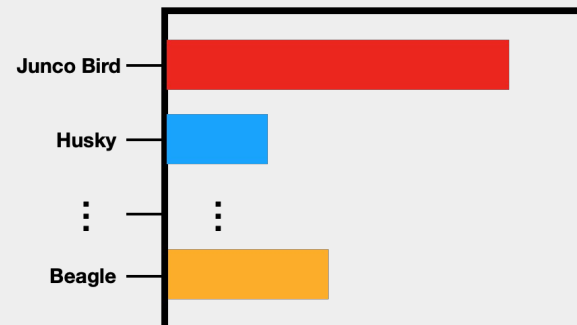
Model



SmoothGrad

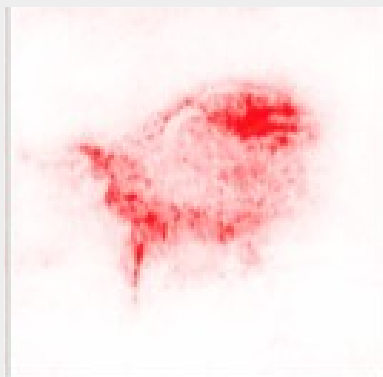


Model



$$\frac{1}{N} \sum_i^N \nabla_{(x+\epsilon)} F_i(x + \epsilon)$$

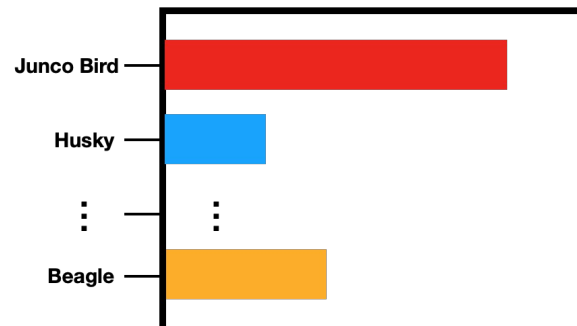

Gaussian noise



Guided Backprop: “Modified Backprop”



Model



activation:

$$f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$$

backpropagation:

$$R_i^l = (f_i^l > 0) \cdot R_i^{l+1}, \text{ where } R_i^{l+1} = \frac{\partial f^{out}}{\partial f_i^{l+1}}$$

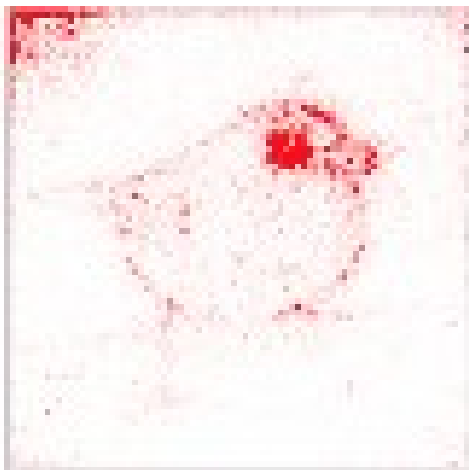
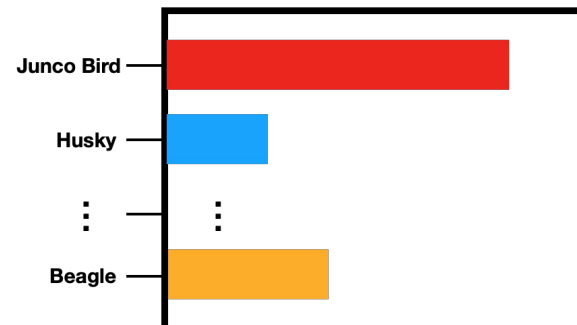
guided
backpropagation:

$$R_i^l = (f_i^l > 0) \cdot (R_i^{l+1} > 0) \cdot R_i^{l+1}$$

Guided Backprop



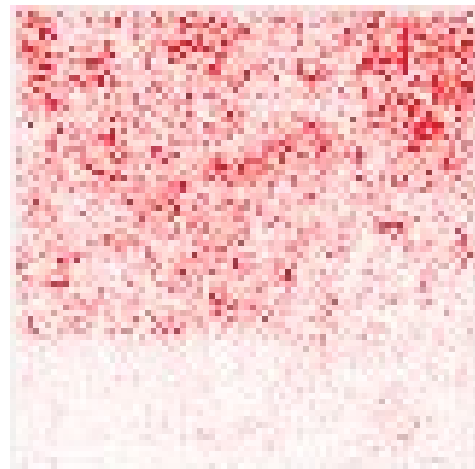
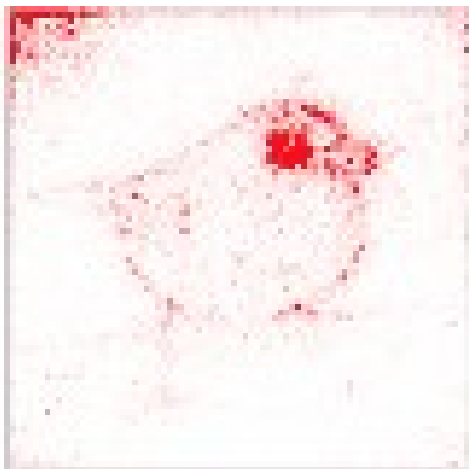
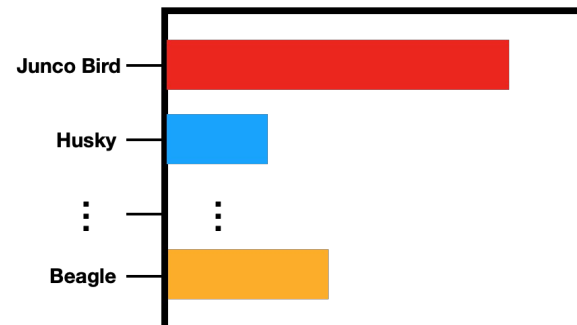
Model



Guided Backprop



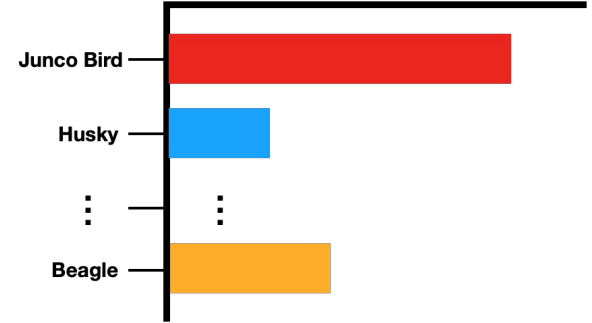
Model



Recap



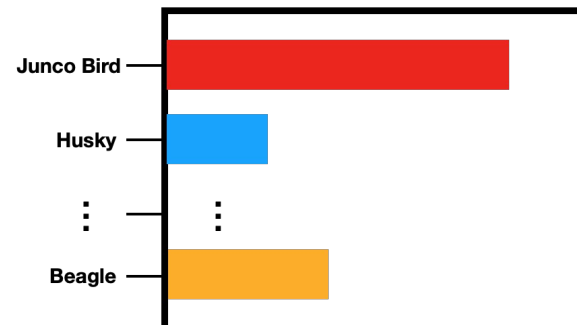
Model



Recap



Model



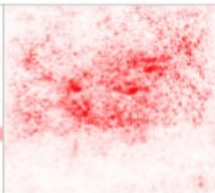
LIME



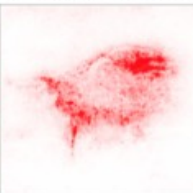
SHAP



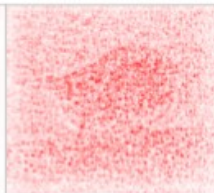
Gradient



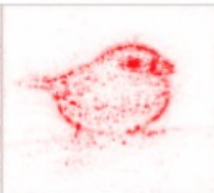
SmoothGrad



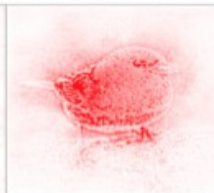
DeConvNet



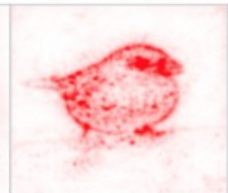
Guided BackProp



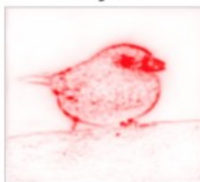
PatternNet



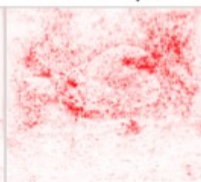
Pattern Attribution



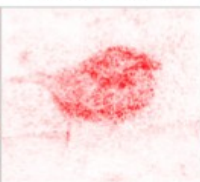
Deep Taylor



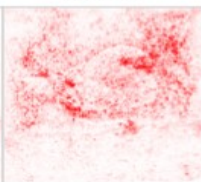
Grad-Input



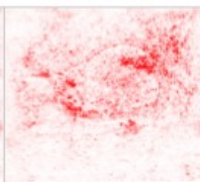
Integrated Gradients



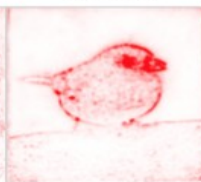
LRP-Z



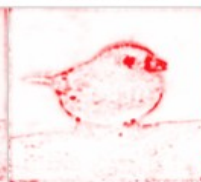
LRP-EPS



LRP-PA



LRP-PB

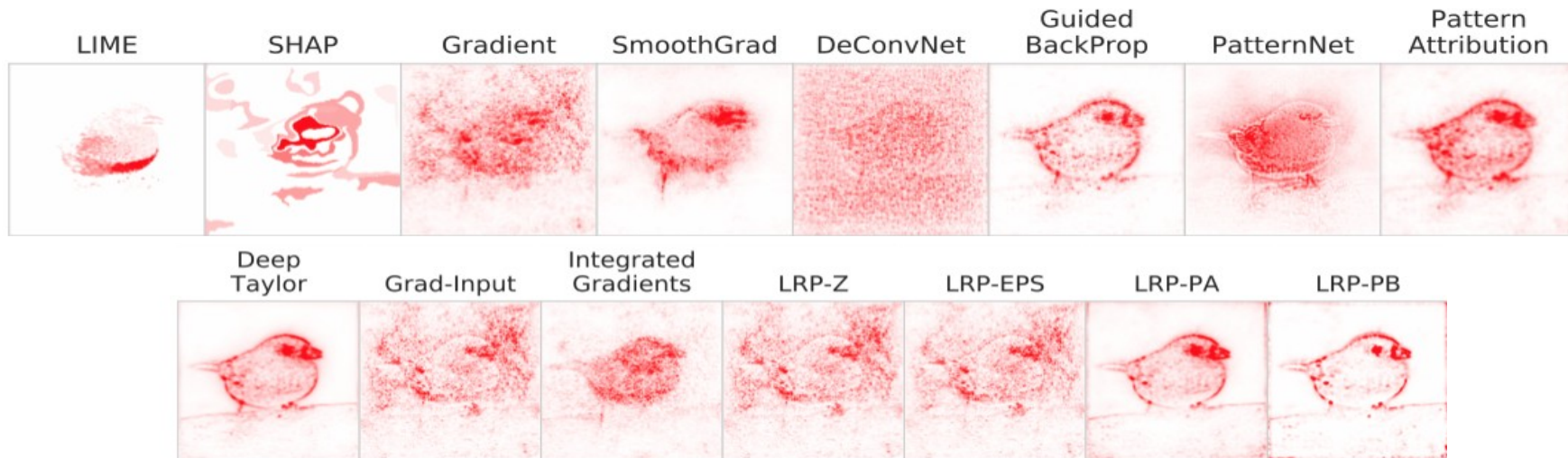
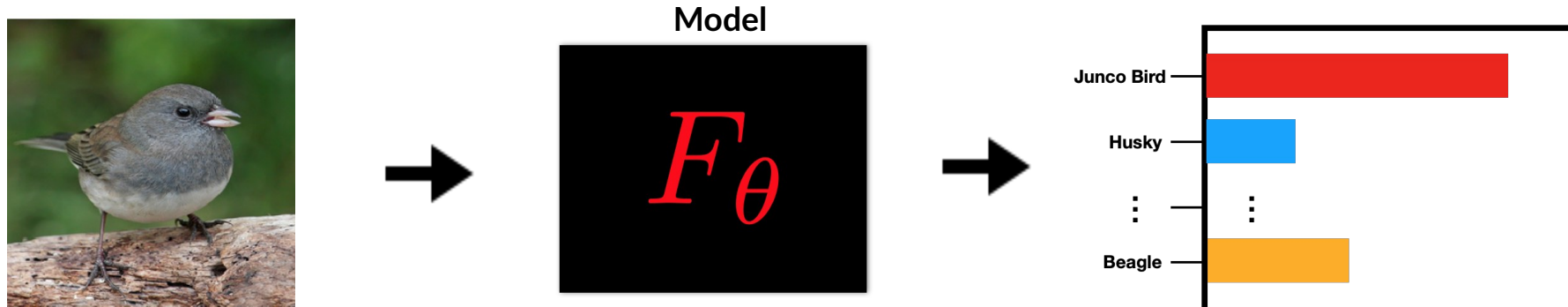


Recap

- **Class Activation Mapping** (Zhou et. al. 2016).
- **Meaningful Perturbation** (Fong et. al. 2017).
- **RISE** (Petsuik et. al. 2018).
- **Extremal Perturbations** (Fong & Patrick 2019).
- **DeepLift** (Shrikumar et. al. 2018).
- **Expected Gradients** (Erion et. al. 2019)
- **Excitation Backprop** (Zhang et. al. 2016)
- **GradCAM** (Selvaraju et. al. 2016)
- **Guided GradCAM** (Selvaraju et. al. 2016)
- **Occlusion** (Zeiler et. al. 2014).
- **Prediction Difference Analysis** (Gu. et. al. 2019).
- **Internal Influence** (Leino et. al. 2018).

See for additional methods: [Samek & Montavon et. al. 2020](#)

Recap: which method should you use?



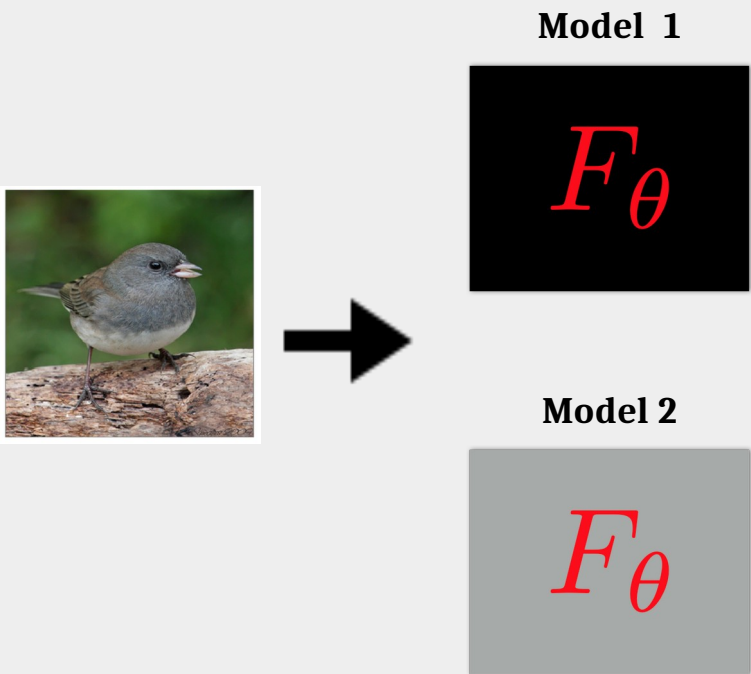
‘Sanity Checks’

Intuitive ‘principles’ that an attribution method should satisfy.

- **‘Model’ Faithfulness:** is the ‘explanation’ sensitive to model parameters?
 - Test: change the model weights and measure corresponding change in explanation.
 - Operationalize by reinitialization of model weights.
- **Data Faithfulness:** is the attribution sensitive to training data?
 - Test: change training label and measure corresponding change in explanation.
 - Operationalize by randomization labelling in training data.

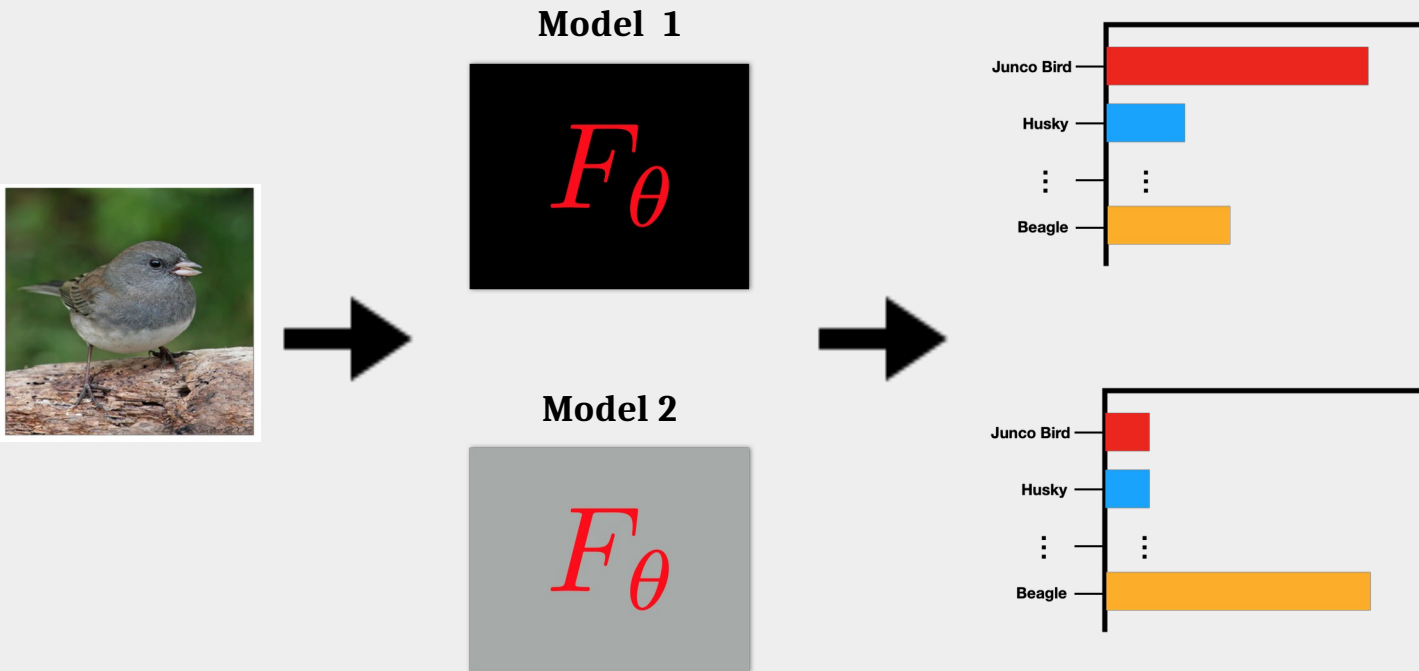
Sensitivity to model parameters

If the parameter settings change of model changes the saliency map should change.



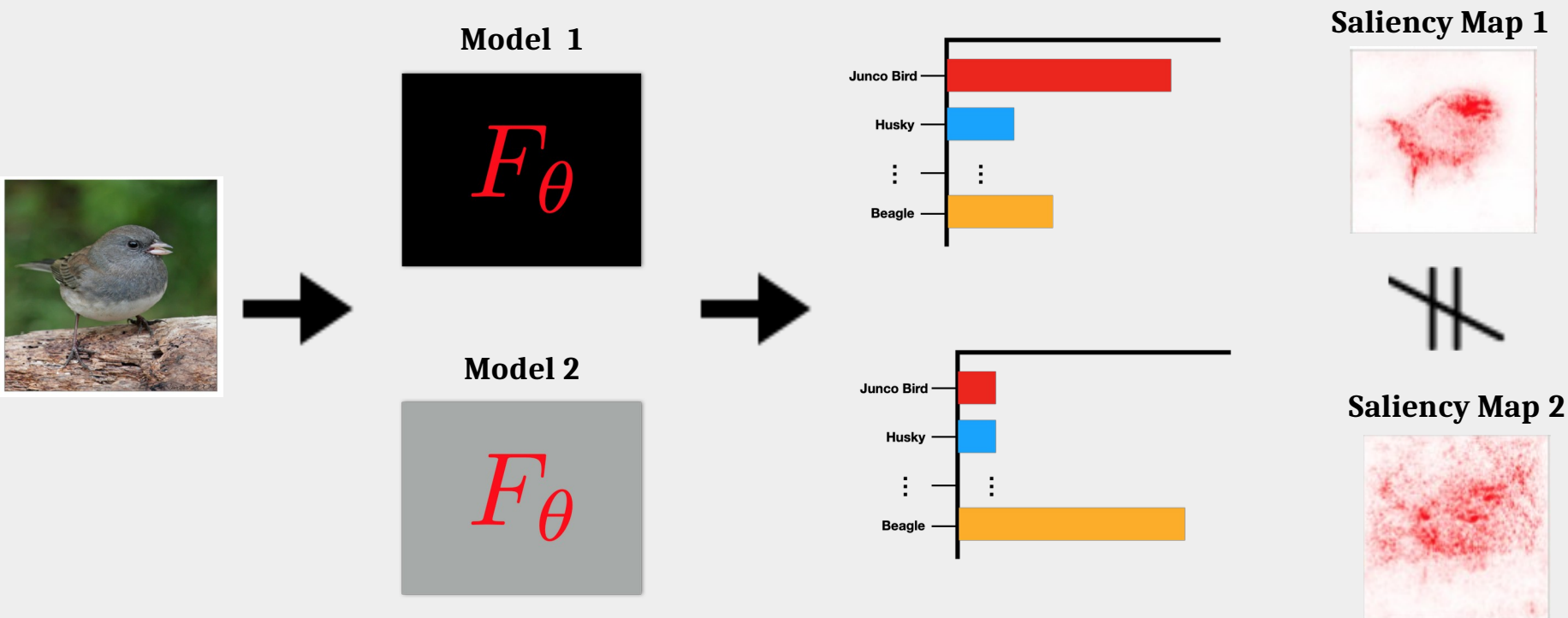
Sensitivity to model parameters

If the parameter settings change of model changes the saliency map should change.

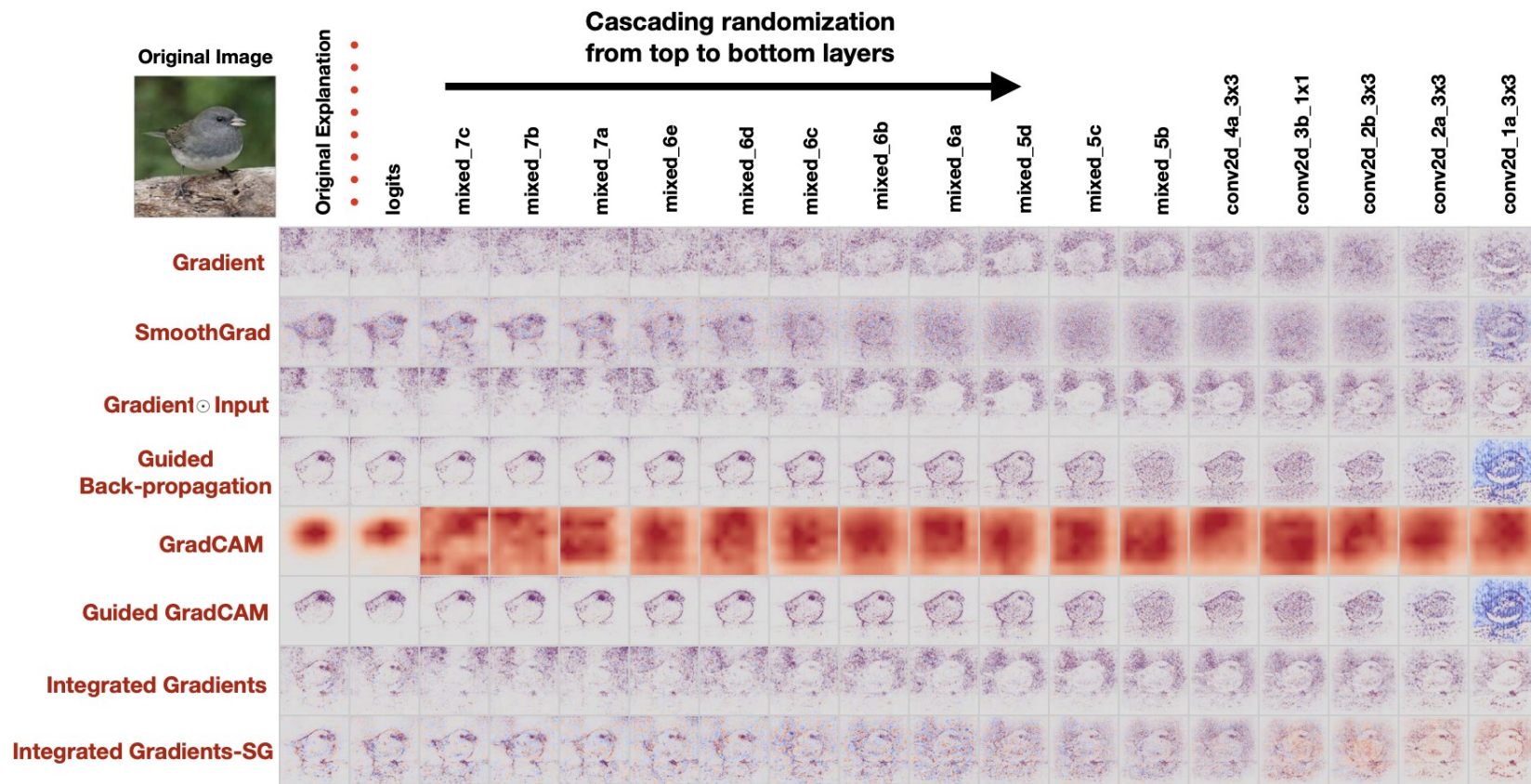


Sensitivity to model parameters

If the parameter settings change of model changes the saliency map should change.

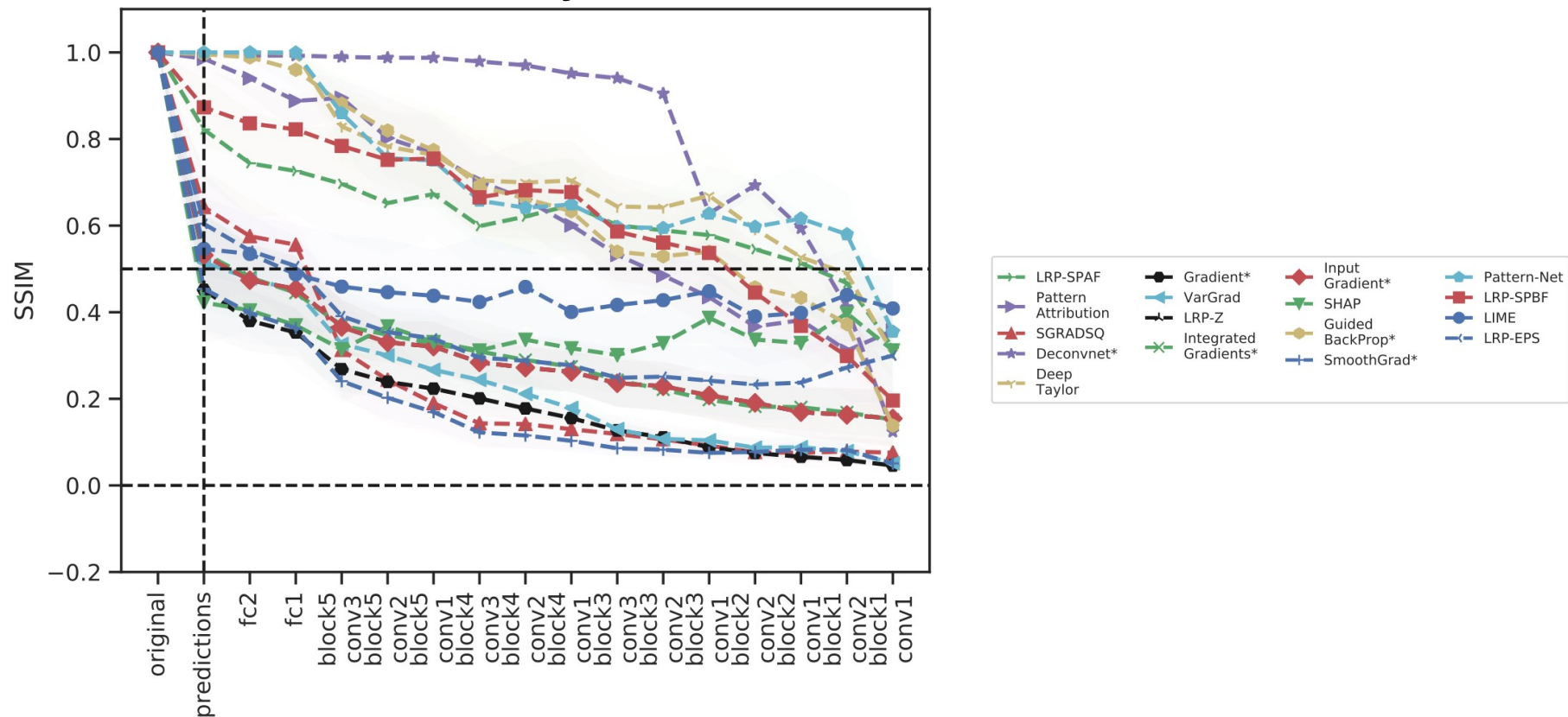


Sensitivity to model parameters

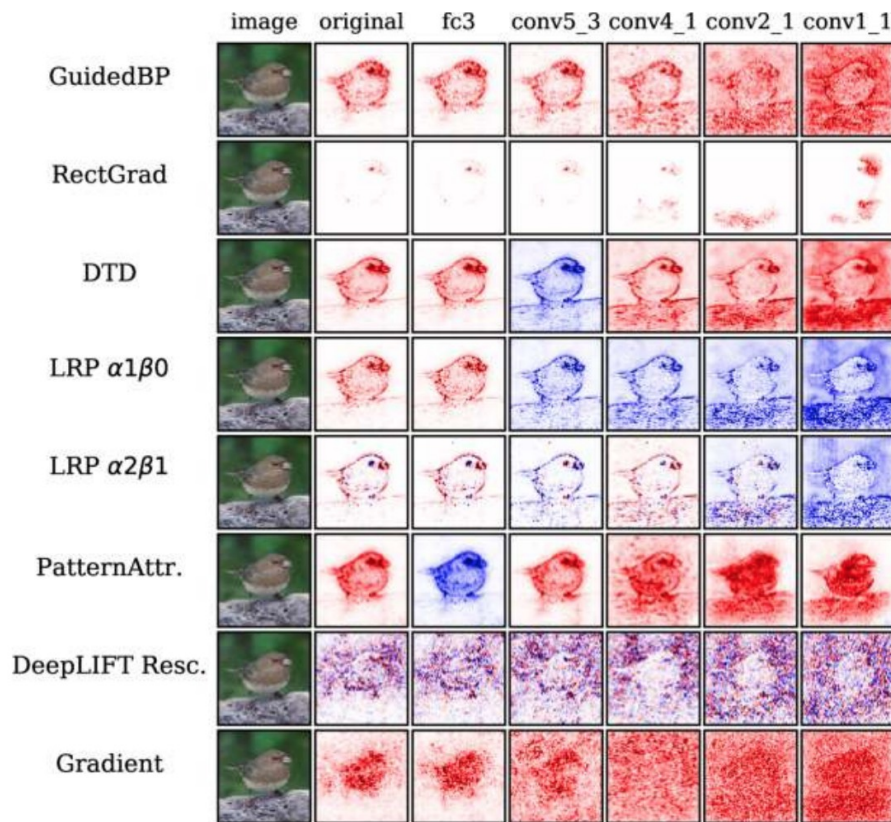


Sensitivity to model parameters

Structural Similarity Index Measure



Modified BackProp Approaches

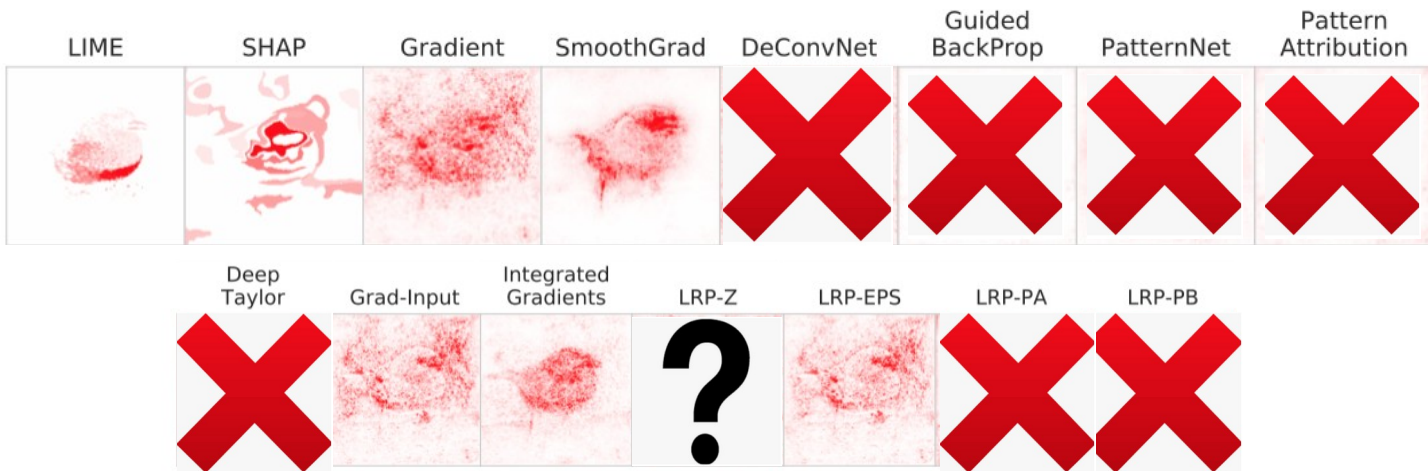
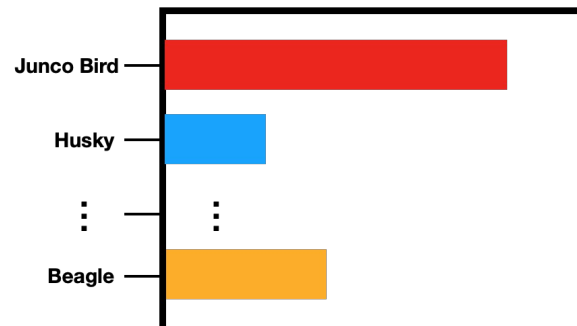


These modified backprop methods **converge to a rank-1 matrix!** This is because the product of a sequence of non-negative matrices (non-orthogonal columns) converges to a rank-1 matrix ([Theorem 1 in Sixt et. al. 2020](#)).

Recap: which method should you use?



Model



Some Takeaways

- Identified certain classes of feature attribution methods that are invariant to higher layer weights.
- ‘Sanity Checks’ are actually ‘weak’ requirements, i.e., does not tell you whether a method is effective.

Some objections

Causal reframing suggests that sanity checks results might be task specific.

Revisiting Sanity Checks for Saliency Maps

Gal Yona
Weizmann Institute of Science

Daniel Greenfeld
Jethor Energy Research

On the Relationship Between Explanation and Prediction: A Causal View

Amir-Hossein Karimi^{1 2 3} **Krikamol Muandet**⁴ **Simon Kornblith**³ **Bernhard Schölkopf**¹ **Been Kim**³

Some objections

Where you choose to perform randomization matters, and perhaps the weight randomization is not the best approach.

Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations

Alexander Binder^{1,2}[0000-0001-9605-6209], Leander Weber³, Sebastian Lapuschkin³[0000-0002-0762-7258],
Grégoire Montavon^{4,5}[0000-0001-7243-6186], Klaus-Robert Müller^{5,6,7,8}, and Wojciech
Samek^{3,5,6}[0000-0002-6283-3265]

More recent observations: Spurious

Beyond faithfulness, it is unclear whether these feature attribution methods are effective for model debugging.

Do Feature Attribution Methods Correctly Attribute Features?

Yilun Zhou¹, Serena Booth¹, Marco Tulio Ribeiro², Julie Shah¹

¹MIT CSAIL, ²Microsoft Research

¹{yilun, serenabooth, julie_a_shah}@csail.mit.edu, ²marcotcr@microsoft.com

More recent observations

Do Input Gradients Highlight Discriminative Features?

RETHINKING THE ROLE OF GRADIENT-BASED ATTRIBUTION METHODS FOR MODEL INTERPRETABILITY

Harshay Shah*

Microsoft Research India
harshay@google.com

Prateek Jain*

Microsoft Research India
prajain@google.com

Praneeth Netrapalli*

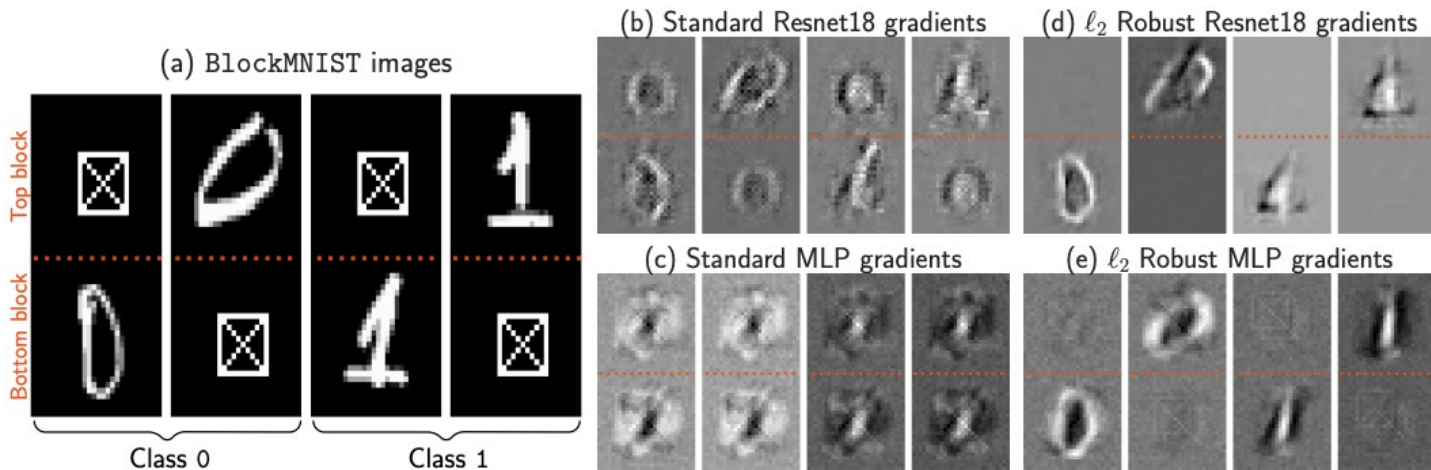
Microsoft Research India
pnetrapalli@google.com

Suraj Srinivas

Idiap Research Institute & EPFL
suraj.srinivas@idiap.ch

François Fleuret

University of Geneva
francois.fleuret@unige.ch



Parting Thoughts

Feature attribution is still important for applications, however, additional is needed to characterize the properties of DNN model training that will result in 'gradients' that capture discriminative signals.

LIME

SHAP

IG

GradCAM

SmoothGrad

**ML
Developer
and
Practitioner**

Exploring the
Explanation
Landscape



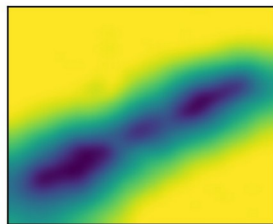
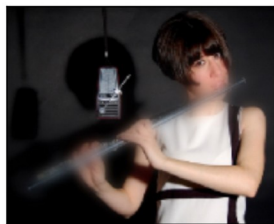
Overview

- ❑ Reliability pillars
- ❑ OpenXAI
- ❑ Is OpenXAI all you need?
- ❑ New directions

flute: 0.9973

flute: 0.0007

Learned Mask



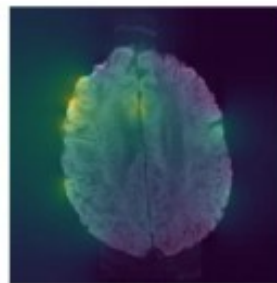
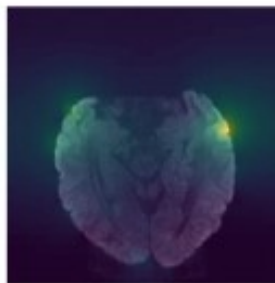
er et al. 2014

ngenberg et al. 2015

works. Sundararajan et al. 2016

ion. Zhou et al. 2016

of any classifier. Ribeiro et al. 2016



Smooth

MP: Inte

Natural images

Fong et al. 2017

nilkov et al. 2017

I Perturbation. Fong et al. 2017

MRI brain scans

Agarwal et al. 2021

SHAP: A Unified Approach to Interpreting Model Predictions. Lundberg et al. 2017

PDA: Visualizing deep neural network decisions: Prediction difference analysis. Zinterhof et al. 2017

From: johnchad@triton.unm.edu (jchadwic)

Subject: Another request for Darwin Fish

Organization: University of New Mexico, Albuquerque

Lines: 11

NNTP-Posting-Host: triton.unm.edu

FIDO: Explaining model generation. Chang et al. 2019

Text

Ribeiro et al. 2016

Expected Gradients: Learning Explainable Models Using Attribution Priors. Erion et al. 2019

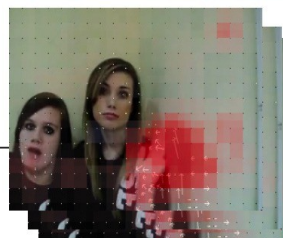
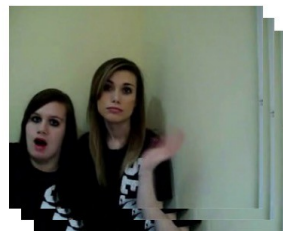
FG-Vis: Interpretable and Fine-Grained Visual Explanations for Classification. Erion et al. 2019

Understanding Deep Networks via Extreme Pruning. Li et al. 2016

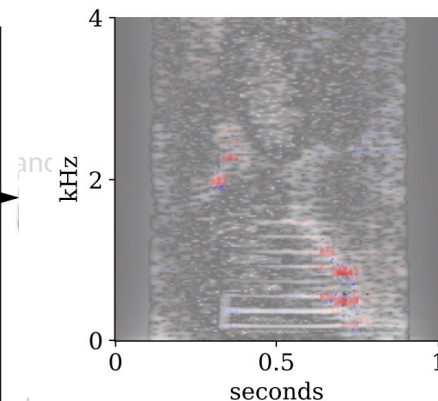
MP-G: Removing input features via a generative model. Srinivasan et al. 2017

Videos

Video From 'wave'



Heatmap



Audio

Becker et al. 2019

Chest X-ray

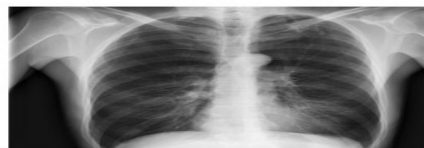
Rajpurkar et al. 2017



Input
Chest X-Ray Image

CheXNet
121-layer CNN

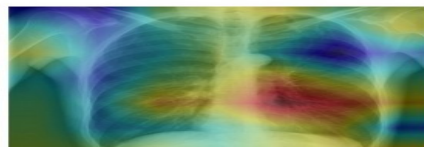
Output
Pneumonia Positive (85%)



Input
Chest X-Ray Image

CheXNet
121-layer CNN

Output
Pneumonia Positive (85%)



et al. 2014

Lundberg et al. 2015

et al. Sundararajan et al. 2018

Zhou et al. 2016

any classifier. Ribeiro et al. 2016

2017

Fong et al. 2017

Lundberg et al. 2017

ysis. Zinterhof et al. 2017



MRI brain scans

Agarwal et al. 2021

Video From 'wave'

4

Detecting biases

Lapuschkin et al. 2016



From: johnchad@triton.unm.edu (johnchad@triton.unm.edu)

Subject: Another request for Darwin

Organization: University of New Mexico

Lines: 11

NNTP-Posting-Host: triton.unm.edu

FIDO: Explain

Text

Ribeiro et al. 2016

Expected Gradient

FG-Vis: Interpretable and Fine-Grained
2019

Understanding Deep Networks via Explanations

MP-G: Removing input features via a generative model

How do we evaluate the reliability of state-of-the-art explanation methods?

Reliability Pillars



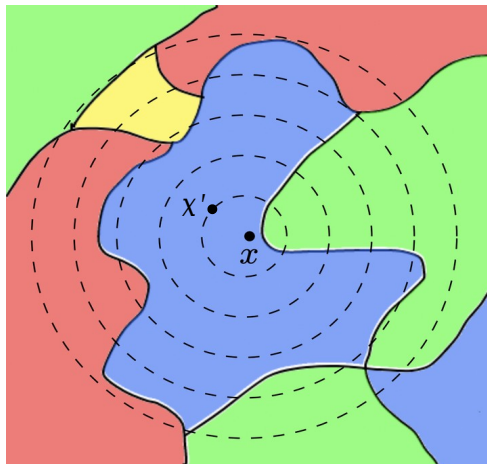
Pillar 1: Faithfulness



Prediction model
to be explained

Masking function

Pillar 2: Stability



Pillar 3: Counterfactual Fairness

-

How do we **pick** an explanation method from the XAI landscape?

- ❑ OpenXAI provides an automated end-to-end pipeline that simplifies and standardizes the evaluation of post hoc explanation methods
- ❑ OpenXAI promotes transparency and reproducibility in benchmarking explanation methods

OpenXAI's Key Components

- ❑ A flexible synthetic data generator and a collection of diverse 7 real-world datasets, 16 pre-trained models, and 6 state-of-the-art explanation methods
- ❑ Open-source implementations of 22 quantitative metrics for evaluating faithfulness, stability (robustness), and fairness of explanation methods
- ❑ First-ever public XAI leaderboards to benchmark explanation methods

XAI ready Dataloaders and Models

```
from openxai import Dataloader
loader_train, loader_test = Dataloader.return_loaders(data_name='german',
download=True)
inputs, labels = iter(loader_test).next()
```

XAI ready Dataloaders and Models

```
from openxai import Dataloader
loader_train, loader_test = Dataloader.return_loaders(data_name='german',
download=True)
inputs, labels = iter(loader_test).next()
```

OpenXAI provides pre-trained models for readily benchmarking explanation methods.

```
from openxai import LoadModel
model = LoadModel(data_name='german', ml_model='ann')
```

OpenXAI Explainers

- ❑ OpenXAI provides ready-to-use implementations of six state-of-the-art feature attribution methods

```
from openxai import Explainer
exp_method = Explainer(method='LIME')
explanations = exp_method.get_explanations(model, X=inputs, y=labels)
```

OpenXAI Explainers

```
@abstractmethod
def get_explanations(self, model, X:
torch.Tensor, y: torch.Tensor):
    """
    Generate explanations for given input/s.
    Parameters
    -----
    model: pre-trained ML model
    X: torch.tensor
        Input in two-dimensional shape (m, n).
    y: torch.tensor
        Labels
    Returns
    -----
    torch.Tensor
        Explanation vector/matrix.
    """
    pass
```

OpenXAI's Evaluation

- OpenXAI provides implementations and ready-to-use APIs for a set of 22 quantitative metrics proposed by prior research to evaluate the faithfulness, stability, and fairness of explanation methods

```
from openxai import Evaluator
metric_evaluator = Evaluator(inputs, labels, model, explanations)
score = metric_evaluator.eval(metric='RIS')
```

OpenXAI's Leaderboard

Explore Leaderboards

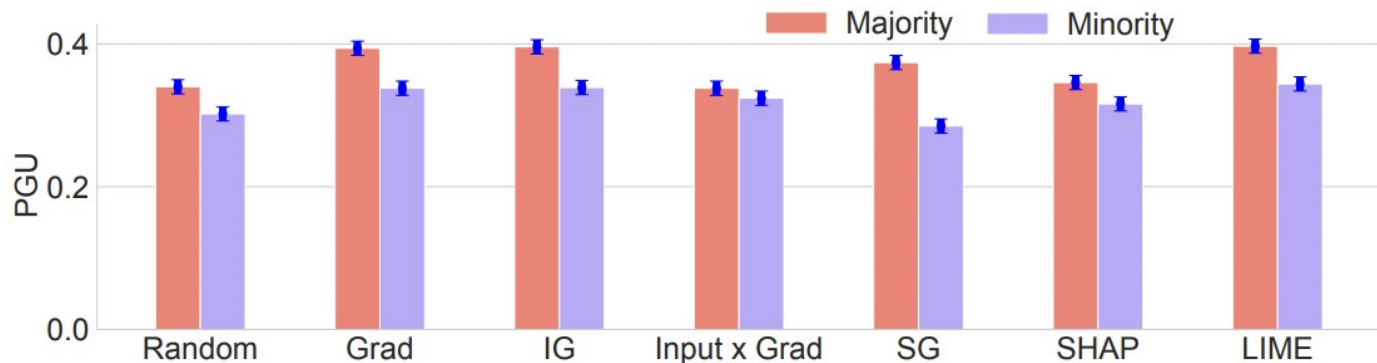
German Credit

Faithfulness

Method	FA ↑	RA	SA	SRA	RC	PRA	PGI	PGU
Vanilla Gradient	0.950	0.950	0.846	0.846	1.000	1.000	0.149	0.174
SmoothGrad	0.950	0.950	0.606	0.606	1.000	1.000	0.202	0.124
Integrated Gradient	0.950	0.950	0.846	0.846	1.000	1.000	0.148	0.173
LIME	0.938	0.810	0.938	0.814	0.998	0.989	0.156	0.169
Gradient x Input	0.785	0.161	0.382	0.071	0.890	0.875	0.171	0.161
SHAP	0.130	0.007	0.112	0.006	-0.053	0.488	0.135	0.180

Exploring the landscape using OpenXAI

- ❑ LIME produces more faithful (+24.9%) explanations
- ❑ Across all real-world datasets, SmoothGrad achieves 63.2% higher RRS values



Is OpenXAI all you need?

- ❑ How to benchmark different non-perturbation-based explanation methods?
- ❑ Benchmarking explanations on other modalities
 - ❑ Vision (Quantus)
 - ❑ NLP (e-ViL)
 - ❑ Graphs (GraphXAI)

New directions

- ❑ Training models using Explanation Feedbacks
- ❑ Differentiable Explainable Curricula for RL Agents
- ❑ Learning Hierarchical and Multi-modal Explanations

Thank you!

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Questions?

